Reaction Time and Accuracy — Revised Analysis

An exploratory report with paired participant summaries

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> **Data and validation.** By default this report regenerates the simulated dataset from `Data_Visualization_RT_example.Rmd`, with seed 2024. To use a supplied CSV instead, set `data_source: "csv"` and update `data_file` in the YAML header. Simulation is an explicit choice, never a fallback for a missing CSV. Numerical checks were run through a Python port of the supplied simulator. Native R execution and knitting could not be tested in the preparation environment; the companion verification report identifies that limitation and documents the calculations actually run.

The required R packages are `dplyr`, `tidyr`, `tibble`, `readr`, `ggplot2`, `patchwork`, `scales`, `viridisLite`, `knitr`, and `rmarkdown`. The last two are also used to knit this file. Install missing packages in the R console before knitting.

Question and context

How do reaction time and response accuracy differ between conditions A and B, and how consistent are the differences across participants?

The supplied reference solution simulates 24 participants, 40 trials in each of two conditions, and two performance outcomes. No actual experiment was conducted to collect these simulated observations. A visual decision task can provide a fictional classroom context, but its recruitment and randomization story must not be presented as the source of the data.

In the simulator, accuracy depends on condition and a participant-specific shift on the **log-odds scale**. Positive RTs are lognormal, with conditional arithmetic means of 480 and 540 ms for correct and incorrect A responses, and 560 and 625 ms in B, plus a participant-specific shift. Thus the incorrect-response mean increase is 60 ms in A and 65 ms in B. The lognormal parameterization subtracts half the log-variance so that the specified value is the conditional arithmetic mean.

The simulator makes B slower and less accurate by construction. It does not simulate learning, fatigue, or dependence on the preceding response. `Trial` is an index from 1 to 40 within each condition; it is not a recorded sequence of an interleaved experiment.

We describe trial distributions and then compare conditions within participants. Our main RT outcome is the median positive, finite RT on correct trials for each participant and condition. Accuracy is calculated from all trials with observed correctness, regardless of RT availability.

Import and check the data

```
if (!params$data_source %in% c("simulation", "csv")) {
  stop("data_source must be simulation or csv.")
}
if (params$data_source == "simulation") {
  RNGkind(kind = "Mersenne-Twister", normal.kind = "Inversion",
          sample.kind = "Rejection")
  set.seed(params$seed)
  n_subj      <- 24
  n_trials    <- 40          # per subject per condition
  conditions  <- c("A (Easy)", "B (Hard)")
```

```r
  # Subject-level random effects
  subj_df <- tibble(
    Subject     = sprintf("S%02d", seq_len(n_subj)),
    re_intercept = rnorm(n_subj, 0, 40),    # random shift in RT (ms)
    re_accuracy  = rnorm(n_subj, 0, 0.05)  # random shift on the log-odds scale
  )

  # Trial-level data
  sim_data <- expand_grid(
    Subject   = subj_df$Subject,
    Condition = conditions,
    Trial     = 1:n_trials
  ) %>%
    left_join(subj_df, by = "Subject") %>%
    mutate(
      # -- Accuracy ----------------------------------------
      # Condition A: p(correct) ~ 0.88 | Condition B: ~ 0.68
      p_correct = case_when(
        Condition == "A (Easy)" ~ plogis( 2.00 + re_accuracy),
        Condition == "B (Hard)" ~ plogis( 0.75 + re_accuracy)
      ),
      Correct = rbinom(n(), 1, p_correct),

      # -- Reaction Time (ms) -------------------------------
      # Base RT: A = 480 ms, B = 560 ms
      # Incorrect-response conditional means are 60 ms higher in A and 65 ms in B
      # Log-normal noise for realistic right-skewed RT
      mu_rt = case_when(
        Condition == "A (Easy)" & Correct == 1 ~ 480,
        Condition == "A (Easy)" & Correct == 0 ~ 540,
        Condition == "B (Hard)" & Correct == 1 ~ 560,
        Condition == "B (Hard)" & Correct == 0 ~ 625
      ),
      RT = rlnorm(n(),
                  meanlog = log(mu_rt + re_intercept) - 0.5 * 0.18^2,
                  sdlog   = 0.18)
    ) %>%
    mutate(
      Correct_f = factor(Correct,
                         levels = c(1, 0),
                         labels = c("Correct", "Incorrect")),
      Condition = factor(Condition, levels = conditions)
    ) %>%
    select(Subject, Condition, Trial, Correct, Correct_f, RT)
  raw <- sim_data
} else {
  if (!file.exists(params$data_file)) {
    stop("CSV not found: ", params$data_file,
         ". Update data_file; observations will not be simulated automatically.")
  }
  raw <- read_csv(
    params$data_file,
    col_types = cols(
```

```
      .default = col_guess(), Subject = col_character(),
      Condition = col_character(), RT = col_double(),
      Correct = col_double(), Correct_f = col_character()
    ), show_col_types = FALSE
  )
  if (nrow(problems(raw)) > 0) {
    print(problems(raw))
    stop("Resolve the CSV parsing problems before analysing the data.")
  }
}
```

```
required <- c("Subject", "Condition", "RT", "Correct", "Correct_f")
if (!all(required %in% names(raw))) {
  stop("Missing columns: ", paste(setdiff(required, names(raw)), collapse = ", "))
}
if (nrow(raw) == 0) stop("The data file contains no observations.")
if (any(is.na(raw$Subject) | trimws(raw$Subject) == "")) {
  stop("Missing or empty participant IDs must be resolved.")
}
# The earlier report accepted A/B only; the supplied simulator uses
# A (Easy)/B (Hard). Map only these documented forms, then validate.
condition_map <- c("A" = "A", "B" = "B",
                   "A (Easy)" = "A", "B (Hard)" = "B")
condition_code <- unname(condition_map[as.character(raw$Condition)])
if (anyNA(condition_code)) {
  stop("Unexpected or missing condition label. Accepted: A, B, A (Easy), B (Hard).")
}
if (!is.numeric(raw$RT) || !is.numeric(raw$Correct)) {
  stop("RT and Correct must be numeric.")
}
if (any(!is.na(raw$Correct) & !raw$Correct %in% c(0, 1))) {
  stop("Correct must be numeric 0/1 or missing.")
}

kable(
  count(raw, Correct, Correct_f, name = "Trials"),
  caption = "Check that numeric correctness and text labels agree."
)
```

Table 1: Check that numeric correctness and text labels agree.

Correct	Correct_f	Trials
0	Incorrect	410
1	Correct	1510

```
expected_response <- ifelse(raw$Correct == 1, "correct", "incorrect")
observed_response <- tolower(trimws(as.character(raw$Correct_f)))
if (any(!is.na(expected_response) & !is.na(observed_response) &
        expected_response != observed_response)) {
  stop("Correct and Correct_f disagree. Resolve the coding before analysis.")
}
```

```r
dati <- raw %>%
  mutate(
    Subject = factor(Subject),
    Condition = factor(condition_code, levels = c("A", "B")),
    Response = factor(Correct, levels = c(1, 0),
                      labels = c("Correct", "Incorrect")),
    valid_rt = is.finite(RT) & RT > 0,
    rt_eligible = valid_rt & coalesce(Correct == 1, FALSE)
  )
```

The analysis assumes that 1 means correct and 0 means incorrect. Confirm this interpretation using the label table and the supplied data documentation. Subsequent calculations use `Correct`, not `Correct_f`. Any label inconsistency needs investigation before interpreting accuracy. Identical recorded rows have not been removed: without a trial identifier, they need not be duplicate records.

Observed design and data quality

```r
audit <- dati %>%
  summarise(
    Trials = n(), Participants = n_distinct(Subject),
    Missing_RT = sum(is.na(RT)),
    Nonfinite_RT = sum(!is.na(RT) & !is.finite(RT)),
    Nonpositive_RT = sum(is.finite(RT) & RT <= 0),
    Missing_correctness = sum(is.na(Correct)),
    Incorrect_responses = sum(Correct == 0, na.rm = TRUE),
    Eligible_RT = sum(rt_eligible)
  )
kable(pivot_longer(audit, everything(), names_to = "Check", values_to = "Count"),
      caption = "Data audit. RT issues and correctness issues can overlap.")
```

Table 2: Data audit. RT issues and correctness issues can overlap.

Check	Count
Trials	1920
Participants	24
Missing_RT	0
Nonfinite_RT	0
Nonpositive_RT	0
Missing_correctness	0
Incorrect_responses	410
Eligible_RT	1510

```r
# Both grouping variables are factors. .drop = FALSE retains absent
# participant-condition combinations with zero trials.
person <- dati %>%
  group_by(Subject, Condition, .drop = FALSE) %>%
  summarise(
    Trials = n(),
    N_scored = sum(!is.na(Correct)),
```

```
    N_missing_score = sum(is.na(Correct)),
    N_correct = sum(Correct == 1, na.rm = TRUE),
    N_RT = sum(rt_eligible),
    Accuracy = safe_ratio(N_correct, N_scored),
    Median_RT = safe_stat(RT[rt_eligible], median),
    All_response_median_RT = safe_stat(RT[valid_rt], median),
    .groups = "drop"
  )

kable(
  person %>% select(Subject, Condition, Trials, N_scored, N_missing_score,
                    N_correct, N_RT),
  caption = paste(
    "Participant-condition audit. N_scored is the accuracy denominator;",
    "N_RT counts eligible correct-response RTs. Zero-trial combinations are retained."
  )
)
```

Table 3: Participant-condition audit. N_scored is the accuracy denominator; N_RT counts eligible correct-response RTs. Zero-trial combinations are retained.

Subject	Condition	Trials	N_scored	N_missing_score	N_correct	N_RT
S01	A	40	40	0	35	35
S01	B	40	40	0	28	28
S02	A	40	40	0	33	33
S02	B	40	40	0	29	29
S03	A	40	40	0	34	34
S03	B	40	40	0	26	26
S04	A	40	40	0	33	33
S04	B	40	40	0	24	24
S05	A	40	40	0	33	33
S05	B	40	40	0	25	25
S06	A	40	40	0	37	37
S06	B	40	40	0	29	29
S07	A	40	40	0	34	34
S07	B	40	40	0	28	28
S08	A	40	40	0	37	37
S08	B	40	40	0	27	27
S09	A	40	40	0	36	36
S09	B	40	40	0	30	30
S10	A	40	40	0	37	37
S10	B	40	40	0	25	25
S11	A	40	40	0	36	36
S11	B	40	40	0	30	30
S12	A	40	40	0	37	37
S12	B	40	40	0	30	30
S13	A	40	40	0	35	35
S13	B	40	40	0	30	30
S14	A	40	40	0	35	35
S14	B	40	40	0	27	27
S15	A	40	40	0	36	36

Subject	Condition	Trials	N_scored	N_missing_score	N_correct	N_RT
S15	B	40	40	0	23	23
S16	A	40	40	0	38	38
S16	B	40	40	0	29	29
S17	A	40	40	0	35	35
S17	B	40	40	0	28	28
S18	A	40	40	0	38	38
S18	B	40	40	0	27	27
S19	A	40	40	0	35	35
S19	B	40	40	0	24	24
S20	A	40	40	0	36	36
S20	B	40	40	0	31	31
S21	A	40	40	0	31	31
S21	B	40	40	0	28	28
S22	A	40	40	0	33	33
S22	B	40	40	0	28	28
S23	A	40	40	0	33	33
S23	B	40	40	0	28	28
S24	A	40	40	0	39	39
S24	B	40	40	0	30	30

```
design_summary <- person %>%
  count(Condition, Trials, name = "Participants")
kable(design_summary,
      caption = "Observed trial counts per participant and condition; planned count: 40.")
```

Table 4: Observed trial counts per participant and condition; planned count: 40.

Condition	Trials	Participants
A	40	24
B	40	24

```
if (params$data_source == "simulation") {
  stopifnot(nrow(dati) == 1920L, n_distinct(dati$Subject) == 24L,
            all(person$Trials == 40L))
}
```

The analysis contains **1920 trials** from **24 participants**, compared with the planned 1,920 trials and 24 participants. There are **0 participant–condition combinations** whose trial counts differ from 40, including absent combinations with zero trials. Participants entirely absent from the file cannot be identified by this audit.

Missing, non-finite, and non-positive RTs do not enter RT summaries. Other positive finite times are retained, including extreme values, because no verified protocol justifies a further timing cutoff. This rule does not certify their plausibility. Missing correctness is excluded from accuracy denominators and is not recoded as an error. That choice may still be affected by selective missingness.

Trial-level reaction-time distributions

These summaries pool eligible correct-response trials within each condition. Participants with more eligible trials contribute more observations. They describe the trial distribution, rather than an equally weighted distribution of participant summaries.

```
rt_summary <- dati %>%
  group_by(Condition, .drop = FALSE) %>%
  summarise(
    Trials = sum(rt_eligible),
    Participants = n_distinct(Subject[rt_eligible]),
    Mean_ms = safe_stat(RT[rt_eligible], mean),
    Median_ms = safe_stat(RT[rt_eligible], median),
    SD_ms = safe_stat(RT[rt_eligible], sd),
    IQR_ms = safe_stat(RT[rt_eligible], IQR),
    .groups = "drop"
  )
kable(rt_summary, digits = 1,
      caption = "Pooled eligible correct-response RTs, in milliseconds.")
```

Table 5: Pooled eligible correct-response RTs, in milliseconds.

Condition	Trials	Participants	Mean_ms	Median_ms	SD_ms	IQR_ms
A	846	24	476.2	466.9	96.8	129.2
B	664	24	554.5	550.5	108.8	141.2

```
rt_trials <- filter(dati, rt_eligible)
histogram_binwidth <- 50

if (nrow(rt_trials) > 0) {
  ggplot(rt_trials, aes(RT)) +
    geom_histogram(
      aes(y = after_stat(density)),
      binwidth = histogram_binwidth, boundary = 0,
      fill = "#0072B2", colour = "white"
    ) +
    facet_wrap(vars(Condition), drop = FALSE,
               labeller = labeller(Condition = condition_labels)) +
    labs(x = "Reaction time (ms)", y = "Density",
         title = "Reaction times on correct trials",
         subtitle = "Positive finite RTs; bin width = 50 ms")
} else {
  cat("No eligible correct-response RTs are available for this plot.")
}
```

The common bin width is a display choice, not an exclusion threshold. Compare widths such as 25 and 100 ms when evaluating whether a visual feature depends on binning. The density scale compares distributional shape, while the table reports the potentially different contributing counts. A long tail can separate the mean from the median; inspect the actual distribution before deciding which best addresses the question.

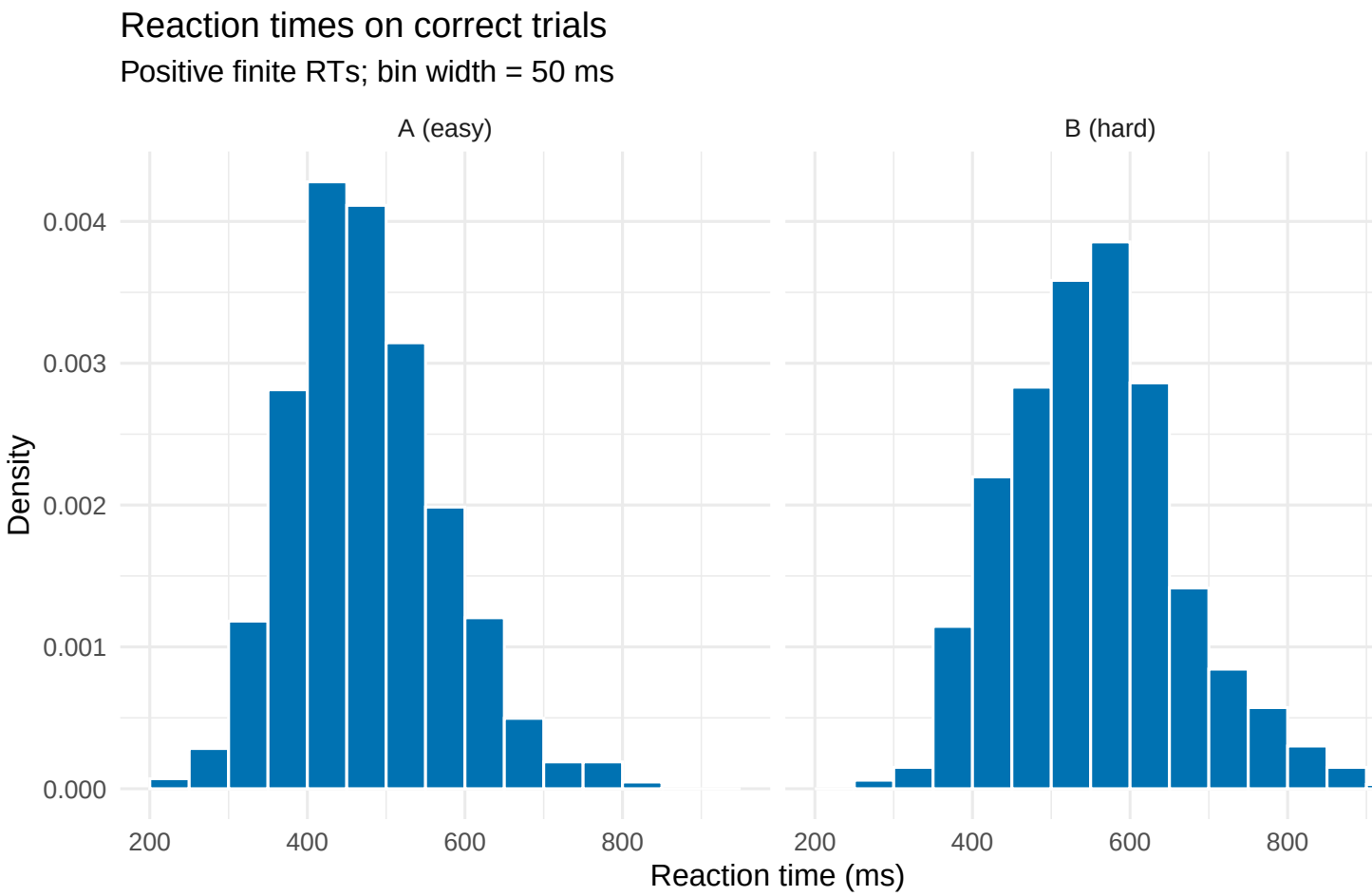


Figure 1: Eligible correct-response RTs pooled within each condition. Each non-empty panel has total histogram area one; panels share axes and 50-ms bin boundaries.

Accuracy and participant-level comparisons

Accuracy denominators and weighting

```
accuracy_summary <- person %>%
  group_by(Condition, .drop = FALSE) %>%
  summarise(
    Participants_with_scores = sum(is.finite(Accuracy)),
    Scored_trials = sum(N_scored),
    Missing_scores = sum(N_missing_score),
    Mean_participant_accuracy_pct = 100 * safe_stat(Accuracy, mean),
    SD_participant_accuracy_pp = 100 * safe_stat(Accuracy, sd),
    Pooled_trial_accuracy_pct = 100 * safe_ratio(sum(N_correct), sum(N_scored)),
    .groups = "drop"
  )
kable(accuracy_summary, digits = 1,
      caption = "Accuracy: equally weighted participant proportions and pooled trial proportions. SD de
```

Table 6: Accuracy: equally weighted participant proportions and pooled trial proportions. SD describes between-participant variation.

Condition	Participants_with_scores	Scored_trials	Missing_scores	Mean_participant_accuracy_pct	SD_participant_accuracy_pp	Pooled_trial_accuracy_pct
A	24	960	0	88.1	5.0	88.1
B	24	960	0	69.2	5.5	69.2

The participant mean gives each contributing participant equal weight. The pooled proportion weights participants by their number of scored trials. The standard deviation is descriptive between-participant variation, not a standard error. These condition-level summaries can include different participants; the following comparisons explicitly retain complete pairs.

Form matched pairs

```
wide <- person %>%
  select(Subject, Condition, Median_RT, Accuracy, All_response_median_RT) %>%
  pivot_wider(names_from = Condition,
              values_from = c(Median_RT, Accuracy, All_response_median_RT))

paired_rt <- wide %>%
  filter(is.finite(Median_RT_A), is.finite(Median_RT_B)) %>%
  mutate(Difference_ms = Median_RT_B - Median_RT_A)
paired_accuracy <- wide %>%
  filter(is.finite(Accuracy_A), is.finite(Accuracy_B)) %>%
  mutate(Difference_pp = 100 * (Accuracy_B - Accuracy_A))

pair_counts <- tibble(
  Outcome = c("Correct-response median RT", "Accuracy"),
  Complete_pairs = c(nrow(paired_rt), nrow(paired_accuracy)),
```

```
  Excluded_participants = nrow(wide) - Complete_pairs
)
kable(pair_counts,
      caption = "Pair availability among participants present in the CSV.")
```

Table 7: Pair availability among participants present in the CSV.

Outcome	Complete_pairs	Excluded_participants
Correct-response median RT	24	0
Accuracy	24	0

Pairing is established by `Subject`, not by the row order of two separately computed vectors. A participant without an eligible value in one condition cannot contribute to that outcome's paired comparison. The audit above shows whether this reflects an absent condition, missing scores, or a lack of eligible correct-response RTs.

```
if (nrow(paired_rt) > 0) {
  rt_long <- paired_rt %>%
    select(Subject, Median_RT_A, Median_RT_B) %>%
    pivot_longer(-Subject, names_to = "Condition",
                 names_prefix = "Median_RT_", values_to = "Median_RT")
  p_paired_rt <- ggplot(rt_long, aes(Condition, Median_RT)) +
    geom_line(aes(group = Subject), colour = "grey65", linewidth = 0.5) +
    geom_point(aes(colour = Condition, shape = Condition), size = 2.5) +
    scale_colour_manual(values = condition_colours) +
    scale_shape_manual(values = c(A = 16, B = 17)) +
    scale_x_discrete(labels = condition_labels) +
    guides(colour = "none", shape = "none") +
    labs(x = NULL, y = "Participant median RT (ms)",
         title = "Reaction time within participants")
  print(p_paired_rt)
} else {
  cat("No complete RT pairs are available; no paired RT plot is produced.")
}
```

```
if (nrow(paired_accuracy) > 0) {
  accuracy_long <- paired_accuracy %>%
    select(Subject, Accuracy_A, Accuracy_B) %>%
    pivot_longer(-Subject, names_to = "Condition",
                 names_prefix = "Accuracy_", values_to = "Accuracy")
  p_paired_accuracy <- ggplot(accuracy_long, aes(Condition, Accuracy)) +
    geom_line(aes(group = Subject), colour = "grey65", linewidth = 0.5) +
    geom_point(aes(colour = Condition, shape = Condition), size = 2.5) +
    scale_colour_manual(values = condition_colours) +
    scale_shape_manual(values = c(A = 16, B = 17)) +
    scale_x_discrete(labels = condition_labels) +
    scale_y_continuous(labels = scales::label_percent(), limits = c(0, 1)) +
    guides(colour = "none", shape = "none") +
    labs(x = NULL, y = "Accuracy", title = "Accuracy within participants")
  print(p_paired_accuracy)
```

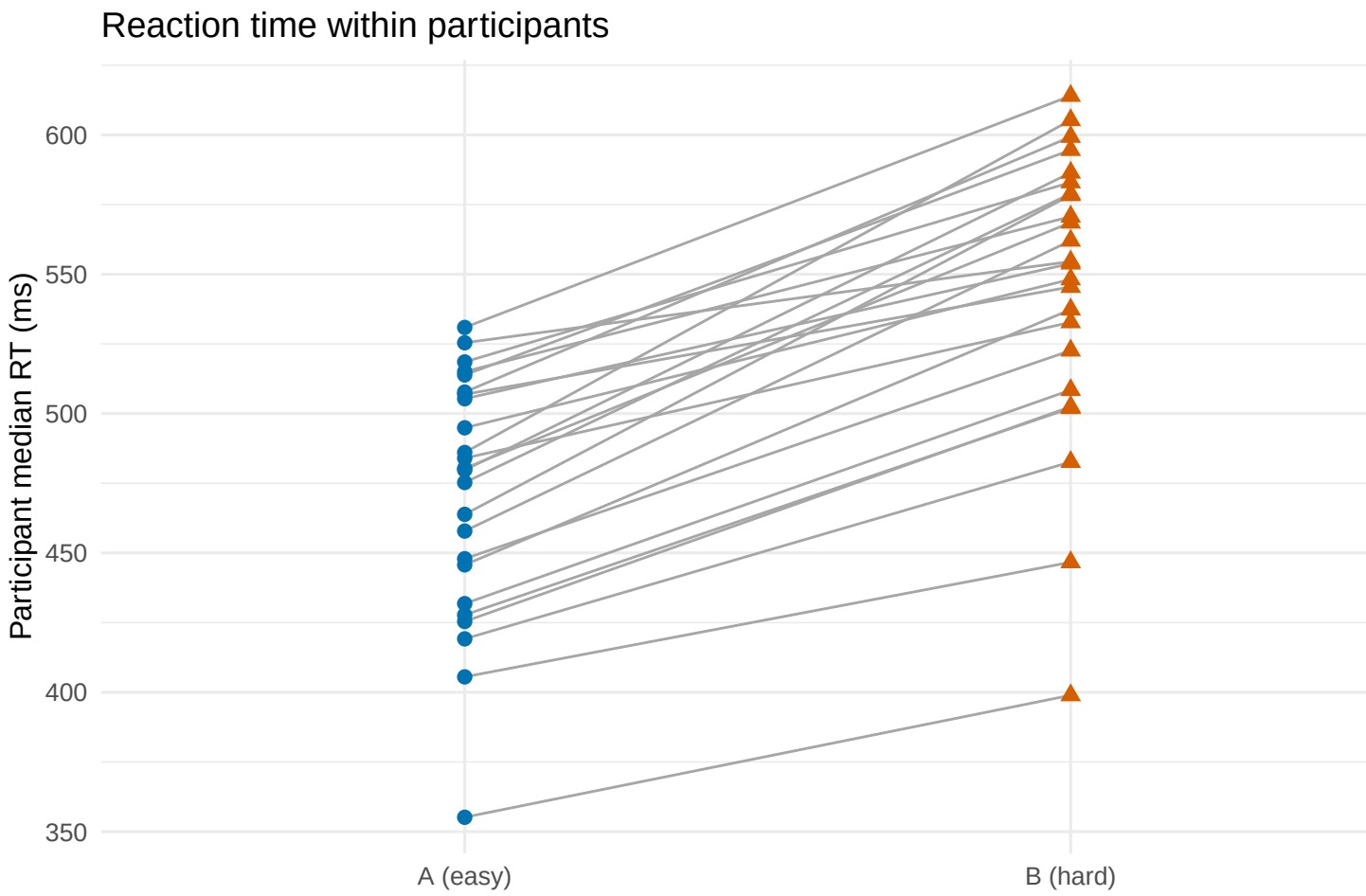


Figure 2: Median eligible correct-response RT for each participant with values in both conditions. Each line connects the same participant; the pair-count table gives the contributing sample size.

```
} else {
  cat("No complete accuracy pairs are available; no paired accuracy plot is produced.")
}
```

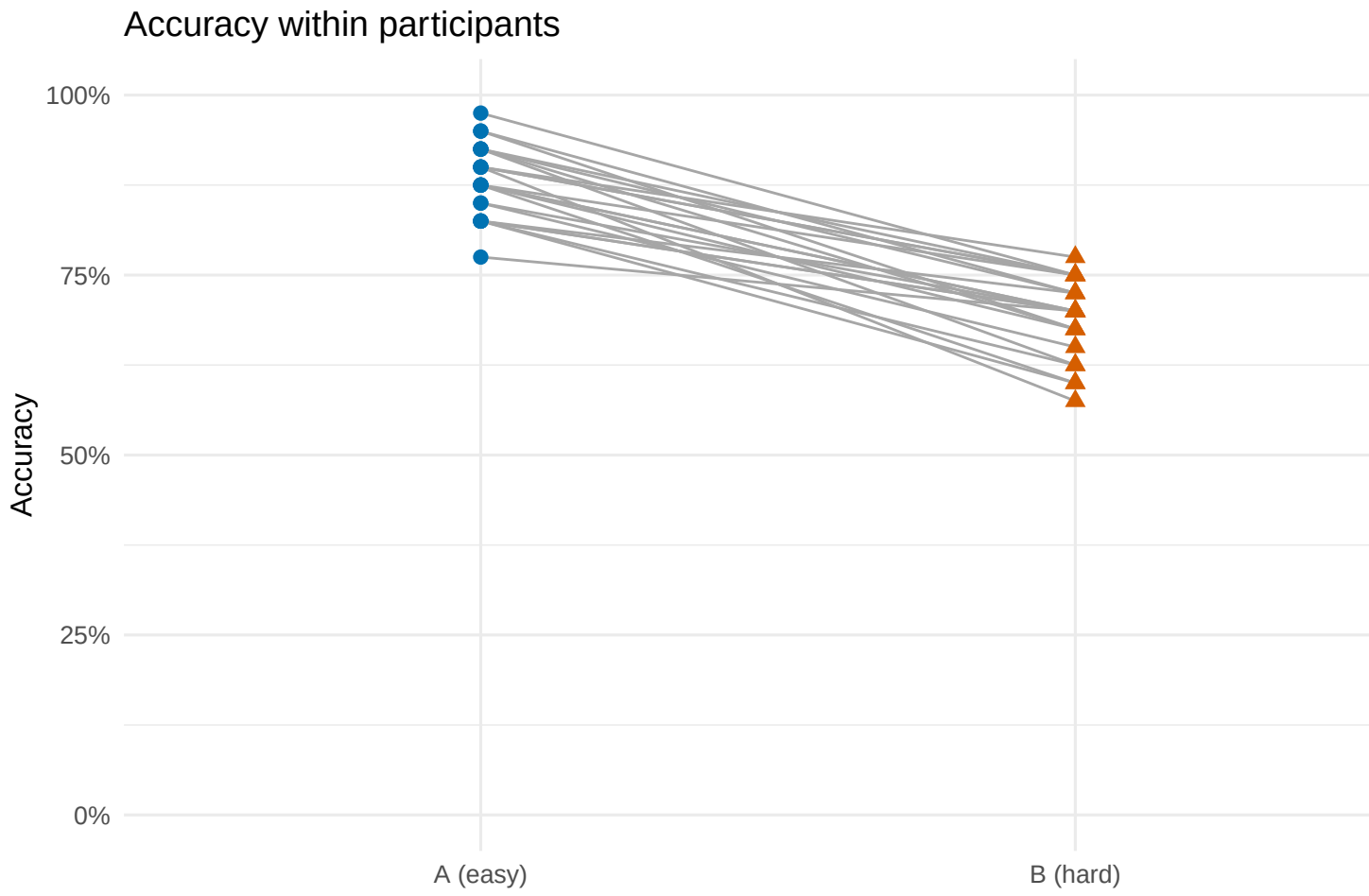


Figure 3: Observed-response accuracy for participants with at least one scored trial in both conditions. Lines preserve participant pairing. RT eligibility does not restrict accuracy denominators.

The connecting lines reveal whether individual differences share a direction. Overlapping points and lines can hide participants, so read the plots together with the counts and numerical summaries. The full 0–100% accuracy scale provides context, although small changes may be less visible.

Within-participant differences

```
differences <- tibble(
  Outcome = c("Participant median RT (ms)", "Accuracy (percentage points)"),
  Pairs = c(nrow(paired_rt), nrow(paired_accuracy)),
  Mean_difference = c(safe_stat(paired_rt$Difference_ms, mean),
                      safe_stat(paired_accuracy$Difference_pp, mean)),
  Median_difference = c(safe_stat(paired_rt$Difference_ms, median),
                        safe_stat(paired_accuracy$Difference_pp, median)),
  SD_difference = c(safe_stat(paired_rt$Difference_ms, sd),
```

```
                    safe_stat(paired_accuracy$Difference_pp, sd)),
  B_greater_than_A = c(sum(paired_rt$Difference_ms > 0),
                       sum(paired_accuracy$Difference_pp > 0)),
  Equal = c(sum(paired_rt$Difference_ms == 0),
            sum(paired_accuracy$Difference_pp == 0)),
  B_less_than_A = c(sum(paired_rt$Difference_ms < 0),
                    sum(paired_accuracy$Difference_pp < 0))
)
kable(differences, digits = 1,
      caption = "B minus A within participants. SD describes variation in paired differences, not uncer
```

Table 8: B minus A within participants. SD describes variation in paired differences, not uncertainty in their mean.

Outcome	Pairs	Mean_difference	Median_difference	SD_difference	B_greater_than_A	Equal	B_less_than_A
Participant median RT (ms)	24	73.9	75.7	25.6	24	0	0
Accuracy (percentage points)	24	-19.0	-18.8	6.5	0	0	24

A positive RT difference means that B has a longer participant median RT; a positive accuracy difference means that B has greater accuracy. Counts above, equal to, and below zero help distinguish a broadly shared pattern from an average driven by a subset of participants.

Across **24 complete RT pairs**, the mean B-minus-A difference in participant median correct-response RT is **73.9 ms**.

Across **24 complete accuracy pairs**, the mean B-minus-A difference in observed-response accuracy is **-19.0 percentage points**.

The first average is an average of **within-participant differences in medians**, not a difference between the pooled trial medians. Both outcomes give equal weight to each contributing paired participant. Their participant sets may differ; assess membership before interpreting the two outcomes jointly as a pattern within the same people.

Compare correct-only and all-response RT descriptions

This comparison retains the same participants as the main paired RT analysis. It replaces each correct-response median with the median of all positive finite RTs, including incorrect responses and responses with missing correctness. It answers a broader question about recorded response times. It is not an alternative way of estimating exactly the same conditional outcome.

```
if (nrow(paired_rt) > 0) {
  subset_comparison <- tibble(
    Response_subset = c("Correct responses", "All responses"),
    Paired_participants = nrow(paired_rt),
    Mean_paired_difference_ms = c(
      mean(paired_rt$Difference_ms),
      mean(paired_rt$All_response_median_RT_B -
             paired_rt$All_response_median_RT_A)
    )
```

```
  )
  kable(subset_comparison, digits = 1,
        caption = "Mean B-minus-A difference in participant medians, using the same paired participants
} else {
  cat("No complete correct-response RT pairs are available for this comparison.")
}
```

Table 9: Mean B-minus-A difference in participant medians, using the same paired participants and positive finite RTs.

Response_subset	Paired_participants	Mean_paired_difference_ms
Correct responses	24	73.9
All responses	24	84.8

Further distributional views

Correct and incorrect responses within each condition

This view separates two questions: how frequent each response is, and how its reaction times are distributed. Percentages in the table use all trials in the corresponding condition as the denominator; RT statistics use positive finite times in the indicated response category. Missing correctness, if present, forms a separate row in the table.

```
response_table <- dati %>%
  group_by(Condition, Response, .drop = FALSE) %>%
  summarise(
    Trials = n(), Valid_RT = sum(valid_rt),
    Mean_ms = safe_stat(RT[valid_rt], mean),
    Median_ms = safe_stat(RT[valid_rt], median),
    SD_ms = safe_stat(RT[valid_rt], sd),
    IQR_ms = safe_stat(RT[valid_rt], IQR), .groups = "drop"
  ) %>%
  group_by(Condition) %>%
  mutate(Within_condition_pct = if (sum(Trials) > 0) {
    100 * Trials / sum(Trials)
  } else NA_real_) %>%
  ungroup() %>%
  select(Condition, Response, Trials, Within_condition_pct, everything())
kable(response_table, digits = 1,
      caption = "Response frequencies within conditions, and valid-time RT summaries in milliseconds.")
```

Table 10: Response frequencies within conditions, and valid-time RT summaries in milliseconds.

Condition	Response	Trials	Within_condition_pct	Valid_RT	Mean_ms	Median_ms	SD_ms	IQR_ms
A	Correct	846	88.1	846	476.2	466.9	96.8	129.2
A	Incorrect	114	11.9	114	546.1	534.1	103.8	141.4
B	Correct	664	69.2	664	554.5	550.5	108.8	141.2
B	Incorrect	296	30.8	296	612.2	603.9	113.7	146.4

```r
density_trials <- dati %>%
  filter(valid_rt, !is.na(Response)) %>%
  group_by(Condition, Response) %>%
  filter(n() >= 2) %>%
  ungroup()
if (nrow(density_trials) > 1) {
  common_bw <- bw.nrd0(density_trials$RT)
  response_medians <- density_trials %>%
    group_by(Condition, Response) %>%
    summarise(Median = median(RT), .groups = "drop")
  p_response_density <- ggplot(
    density_trials, aes(RT, colour = Response, linetype = Response)
  ) +
    geom_density(bw = common_bw, linewidth = 0.9) +
    geom_vline(data = response_medians,
               aes(xintercept = Median, colour = Response, linetype = Response),
               linewidth = 0.5) +
    facet_wrap(vars(Condition), drop = FALSE,
               labeller = labeller(Condition = condition_labels)) +
    scale_colour_manual(values = c(Correct = "#0072B2", Incorrect = "#D55E00")) +
    scale_linetype_manual(values = c(Correct = "solid", Incorrect = "dashed")) +
    labs(x = "Reaction time (ms)", y = "Density",
         title = "Correct and incorrect response times",
         subtitle = sprintf("Shared bandwidth = %.1f ms; vertical lines mark medians", common_bw)) +
    theme(legend.position = "bottom")
  print(p_response_density)
} else {
  cat("Insufficient observations for response-specific density curves.")
}
```

Only groups with at least two usable observations receive a density curve. Changing bandwidth changes apparent smoothness; it does not change the response percentages. The simulator makes incorrect responses slower within each condition. That is a same-trial association, not post-error slowing, which concerns the response following an error.

Boxplots with observed values

```r
if (nrow(rt_trials) > 0) {
  p_rt_box <- ggplot(rt_trials, aes(Condition, RT)) +
    geom_boxplot(width = 0.35, outlier.shape = NA, fill = "grey94") +
    geom_point(aes(colour = Condition, shape = Condition),
               position = position_jitter(width = 0.12, height = 0, seed = 2024),
               alpha = 0.25, size = 1) +
    scale_colour_manual(values = condition_colours) +
    scale_shape_manual(values = c(A = 16, B = 17)) +
    scale_x_discrete(labels = condition_labels) +
    guides(colour = "none", shape = "none") +
    labs(x = NULL, y = "Reaction time (ms)", title = "Correct-response times and their spread")
  print(p_rt_box)
}
```

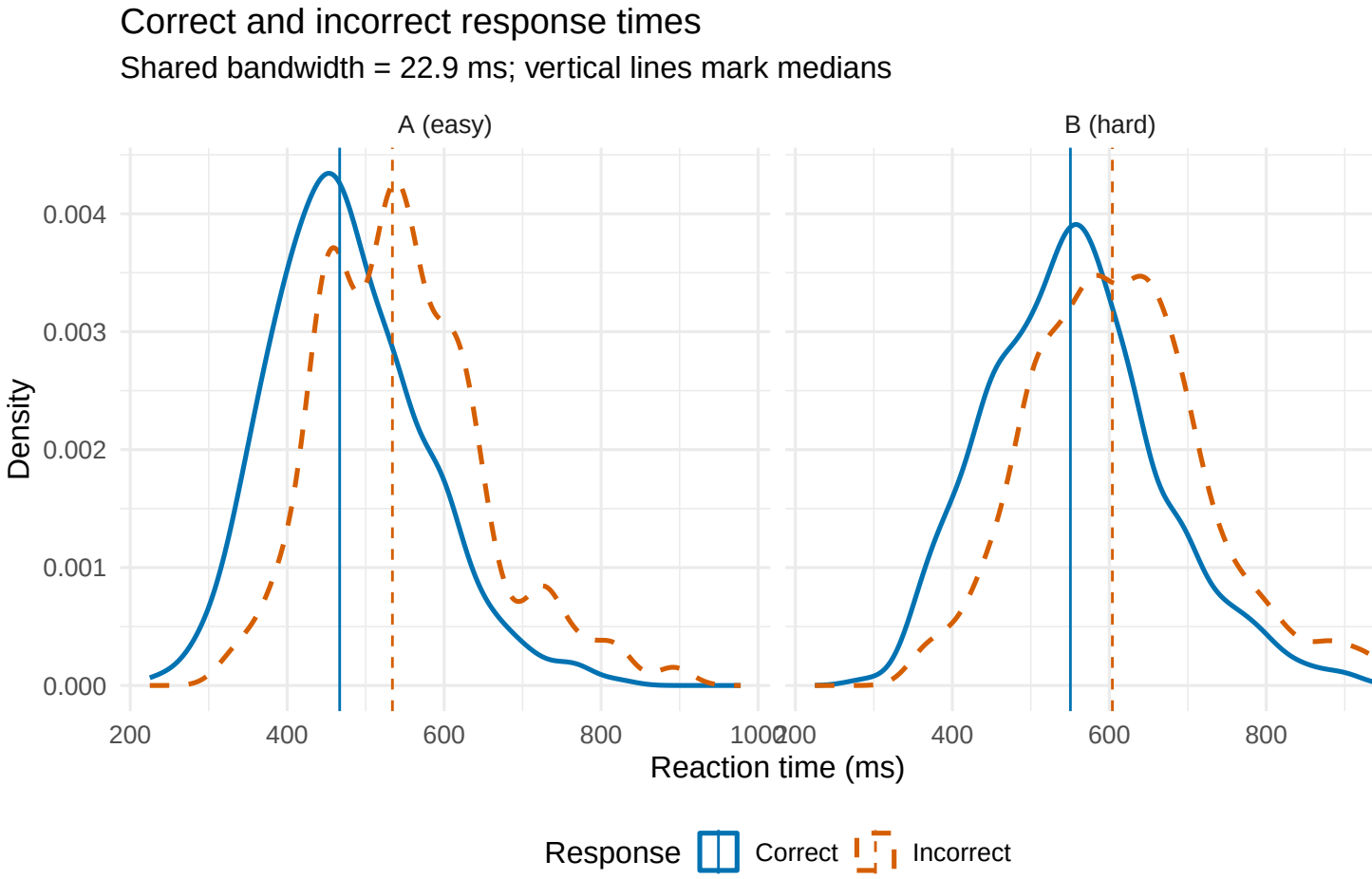


Figure 4: RT distributions conditional on condition and correctness. Curves share one data-derived bandwidth; each curve describes shape, not the frequency of that response.

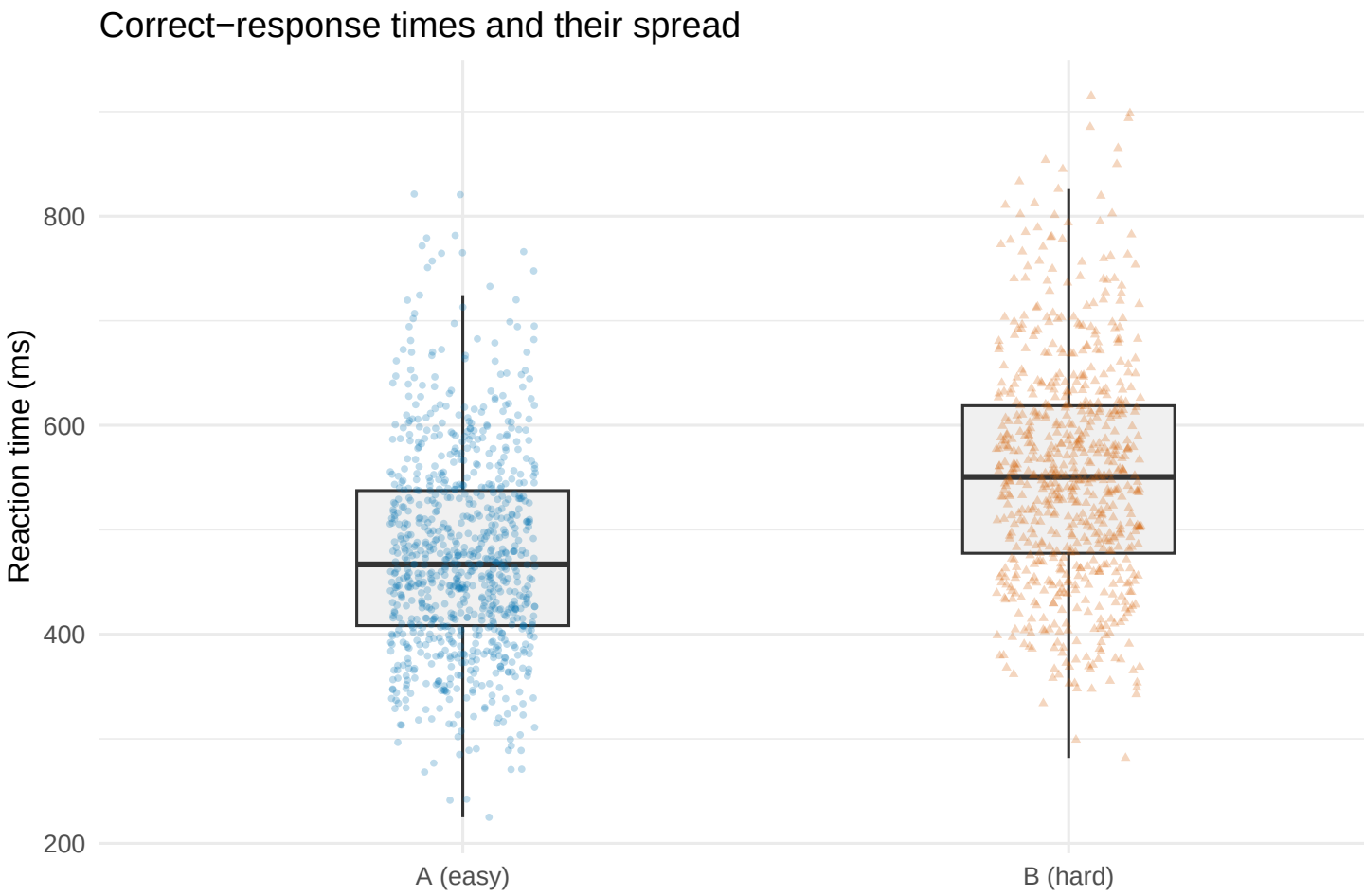


Figure 5: Eligible correct-response RTs. Points are jittered horizontally only; every vertical position remains an observed time. All eligible trials are plotted.

Setting `outlier.shape = NA` suppresses the boxplot's duplicate outlier symbols; it does not remove observations. The point layer still includes all eligible times. These pooled distributions complement the paired participant plots and do not replace them.

Reaction time and accuracy jointly

Both plots below use the same participants with complete values for both outcomes. The left plot retains each participant's A-to-B movement in the two-dimensional outcome space. The right plot shows their paired changes. It helps distinguish changes shared across participants from an association created by pooling the two condition clusters.

```
joint <- wide %>%
  filter(is.finite(Median_RT_A), is.finite(Median_RT_B),
         is.finite(Accuracy_A), is.finite(Accuracy_B)) %>%
  mutate(Delta_RT = Median_RT_B - Median_RT_A,
         Delta_accuracy_pp = 100 * (Accuracy_B - Accuracy_A))

if (nrow(joint) > 0) {
  joint_long <- person %>% semi_join(joint, by = "Subject")
  p_joint_levels <- ggplot() +
    geom_segment(data = joint,
                 aes(x = Accuracy_A, xend = Accuracy_B,
                     y = Median_RT_A, yend = Median_RT_B),
                 colour = "grey70", linewidth = 0.5) +
    geom_point(data = joint_long,
               aes(Accuracy, Median_RT, colour = Condition, shape = Condition),
               size = 2.5) +
    scale_colour_manual(values = condition_colours, labels = condition_labels) +
    scale_shape_manual(values = c(A = 16, B = 17), labels = condition_labels) +
    scale_x_continuous(labels = scales::label_percent()) +
    labs(x = "Accuracy", y = "Participant median RT (ms)",
         title = "Levels in each condition") + theme(legend.position = "bottom")

  p_joint_changes <- ggplot(joint, aes(Delta_accuracy_pp, Delta_RT)) +
    geom_hline(yintercept = 0, colour = "grey65", linetype = "dashed") +
    geom_vline(xintercept = 0, colour = "grey65", linetype = "dashed") +
    geom_point(size = 2.5, colour = "#0072B2") +
    labs(x = "Accuracy difference, B - A (percentage points)",
         y = "Median RT difference, B - A (ms)", title = "Paired changes")
  print(p_joint_levels + p_joint_changes)
} else {
  cat("No participants have complete pairs for both outcomes.")
}
```

There are **24 joint complete pairs**; among them, **24** have both longer median correct-response RTs and lower accuracy in B. This is a description of the observed changes. A speed–accuracy trade-off is not established merely by drawing an RT-versus-accuracy scatter plot.

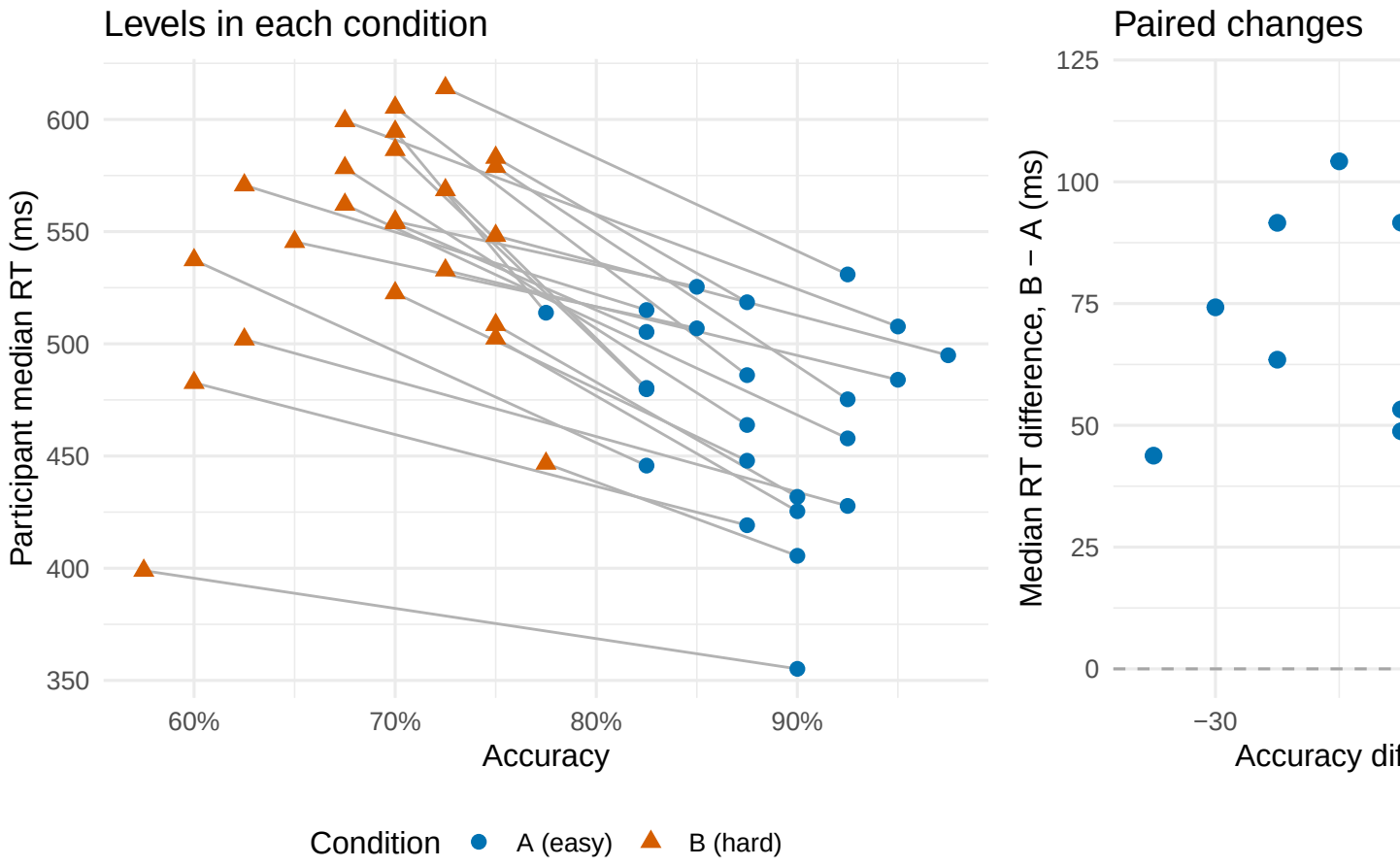


Figure 6: Joint complete pairs only. Segments connect A and B for the same participant. In the changes plot, upper-left means longer RT and lower accuracy in B; lower-right means shorter RT and higher accuracy in B.

Participant overview

The heatmaps share the same participant order, ranked by average observed accuracy over the available conditions. Each colour scale has its own units; colours cannot be compared numerically between the two heatmaps. Missing outcomes are labelled rather than replaced by zero.

```
subject_order <- person %>%
  group_by(Subject) %>%
  summarise(Mean_accuracy = safe_stat(Accuracy, mean), .groups = "drop") %>%
  arrange(Mean_accuracy, Subject) %>% pull(Subject) %>% as.character()
heat <- person %>% mutate(Subject = factor(Subject, levels = subject_order))

# Text contrast follows the actual viridis colour assigned to each value.
contrast_colour <- function(x, limits) {
  z <- if (diff(limits) > 0) (x - limits[1]) / diff(limits) else rep(0.5, length(x))
  pal <- viridisLite::viridis(256)
  colour <- rep("grey85", length(x))
  ok <- is.finite(z)
  colour[ok] <- pal[1 + round(255 * pmax(0, pmin(1, z[ok])))]
  rgb <- t(grDevices::col2rgb(colour)) / 255
  linear <- ifelse(rgb <= 0.04045, rgb / 12.92, ((rgb + 0.055) / 1.055)^2.4)
  luminance <- drop(linear %*% c(0.2126, 0.7152, 0.0722))
  ifelse(luminance > 0.179, "black", "white")
}
finite_medians <- heat$Median_RT[is.finite(heat$Median_RT)]
rt_limits <- if (length(finite_medians) > 0) range(finite_medians) else c(0, 1)
heat <- heat %>% mutate(
  Acc_label = ifelse(is.finite(Accuracy), scales::percent(Accuracy, accuracy = 1), "NA"),
  RT_label = ifelse(is.finite(Median_RT), sprintf("%.0f", Median_RT), "NA"),
  Acc_text = contrast_colour(Accuracy, c(0, 1)),
  RT_text = contrast_colour(Median_RT, rt_limits)
)

p_accuracy_heatmap <- ggplot(heat, aes(Condition, Subject)) +
  geom_tile(aes(fill = Accuracy), colour = "white") +
  geom_text(aes(label = Acc_label, colour = Acc_text), size = 3) +
  scale_colour_identity() +
  scale_fill_viridis_c(limits = c(0, 1), na.value = "grey85",
                       labels = scales::label_percent(), name = "Accuracy") +
  scale_x_discrete(labels = condition_labels) +
  labs(x = NULL, y = "Participant", title = "Accuracy") + theme(panel.grid = element_blank())
p_rt_heatmap <- ggplot(heat, aes(Condition, Subject)) +
  geom_tile(aes(fill = Median_RT), colour = "white") +
  geom_text(aes(label = RT_label, colour = RT_text), size = 3) +
  scale_colour_identity() +
  scale_fill_viridis_c(limits = rt_limits, na.value = "grey85", name = "RT (ms)") +
  scale_x_discrete(labels = condition_labels) +
  labs(x = NULL, y = NULL, title = "Median correct-response RT") +
  theme(panel.grid = element_blank())
print(p_accuracy_heatmap + p_rt_heatmap)
```

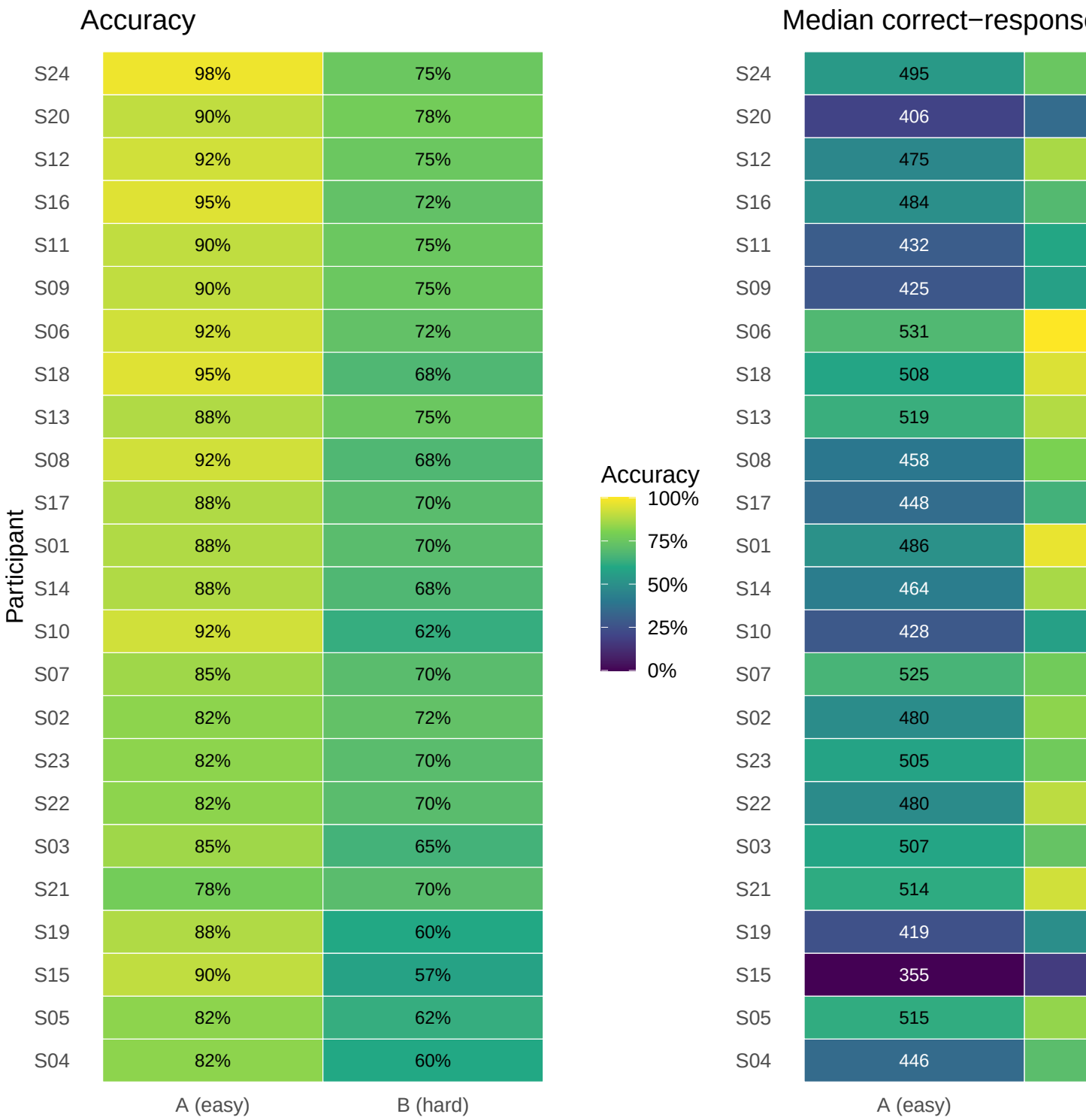


Figure 7: Participant-condition summaries with the same participant ordering in both panels. Accuracy uses observed correctness scores; RT uses eligible correct responses. Labels give rounded numerical values alongside colour.

Supplementary distributional comparisons

Paired RT quantile differences

For each participant retained in the main paired RT comparison, calculate five quantiles of eligible RTs in each condition. Compare the same quantile within each participant, then average those differences. This answers whether the descriptive difference is similar across the distribution. It does not establish a processing mechanism or provide an inferential band.

```
if (nrow(paired_rt) > 0) {
  probabilities <- c(0.1, 0.3, 0.5, 0.7, 0.9)
  quantile_pairs <- rt_trials %>%
    semi_join(paired_rt, by = "Subject") %>%
    group_by(Subject, Condition) %>%
    summarise(
      Probability = list(probabilities),
      RT_quantile = list(quantile(RT, probabilities, names = FALSE, type = 7)),
      .groups = "drop"
    ) %>%
    unnest(c(Probability, RT_quantile)) %>%
    pivot_wider(names_from = Condition, values_from = RT_quantile) %>%
    filter(is.finite(A), is.finite(B))
  delta_summary <- quantile_pairs %>%
    group_by(Probability) %>%
    summarise(Participants = n(), Mean_RT_ms = mean((A + B) / 2),
              Mean_difference_ms = mean(B - A), .groups = "drop") %>%
    arrange(Probability)
  kable(delta_summary, digits = 1, caption = "Paired correct-response quantile comparisons.")
  p_quantile_difference <- ggplot(delta_summary, aes(Mean_RT_ms, Mean_difference_ms)) +
    geom_hline(yintercept = 0, colour = "grey65", linetype = "dashed") +
    geom_path(colour = "#0072B2") + geom_point(size = 2.5, colour = "#0072B2") +
    geom_text(aes(label = scales::percent(Probability)), vjust = -0.7, size = 3) +
    scale_y_continuous(expand = expansion(mult = c(0.05, 0.15))) +
    labs(x = "Average of A and B participant quantiles (ms)",
         y = "Mean paired quantile difference, B - A (ms)",
         title = "Where in the distribution do the conditions differ?")
  print(p_quantile_difference)
} else {
  cat("No complete RT pairs are available for quantile comparisons.")
}
```

Accuracy by relative response speed

This optional view divides scored, positive finite RTs into five rank bins within each participant and condition, then averages participant-bin accuracy with equal participant weight. A participant–condition combination needs at least five eligible observations. Bins describe relative speed within that participant and condition, not common millisecond cutoffs across people.

```
caf_person <- dati %>%
  filter(valid_rt, !is.na(Correct)) %>%
  group_by(Subject, Condition) %>%
  filter(n() >= 5) %>%
```

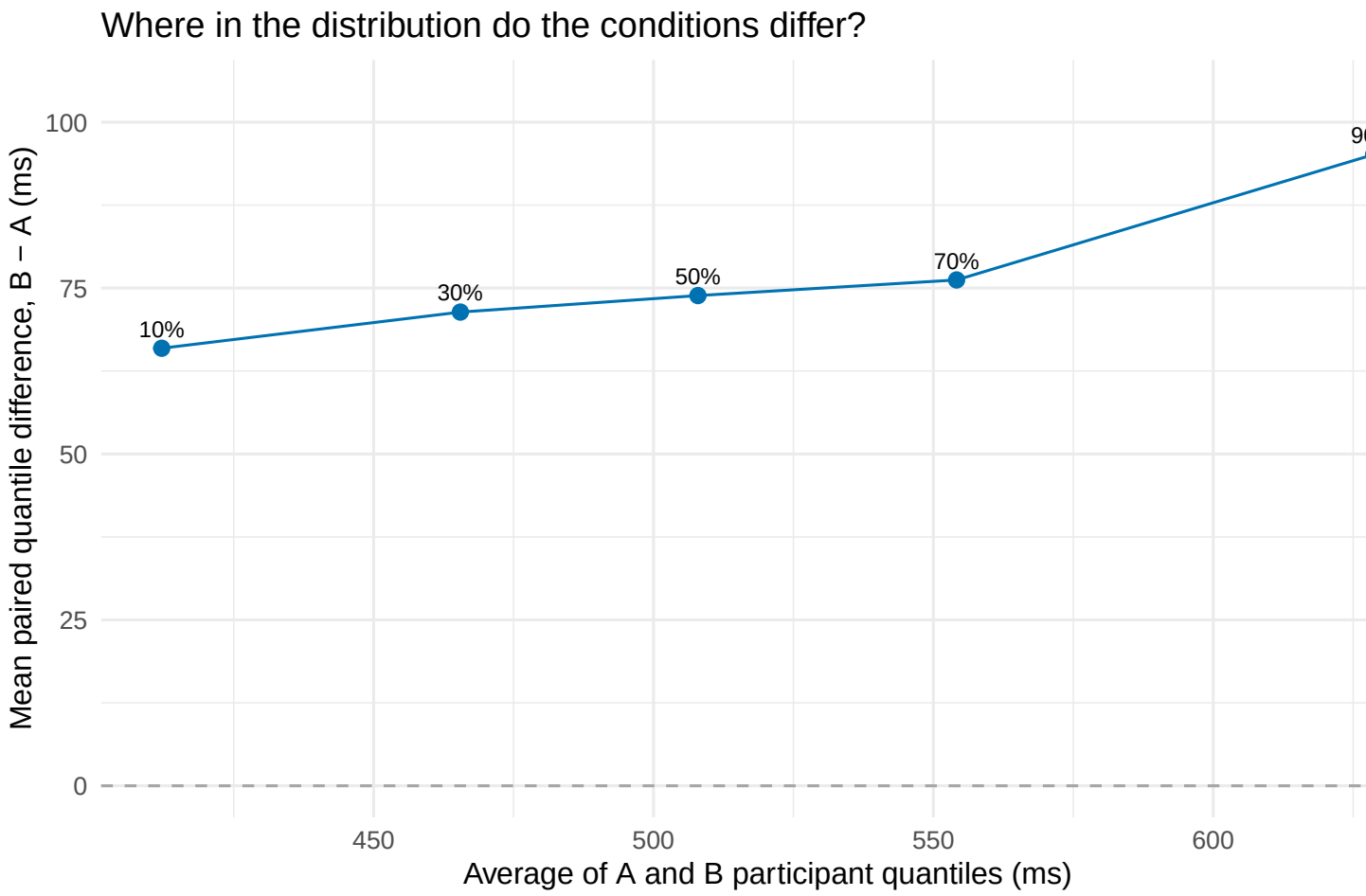


Figure 8: Mean within-participant B-minus-A differences at five RT quantiles. The same paired participants contribute to every quantile. Points are joined in quantile order; no confidence band is implied.

```
  mutate(Quintile = ntile(RT, 5)) %>%
  group_by(Subject, Condition, Quintile) %>%
  summarise(Accuracy = mean(Correct), Trials = n(), .groups = "drop")
if (nrow(caf_person) > 0) {
  caf_summary <- caf_person %>%
    group_by(Condition, Quintile) %>%
    summarise(Participants = n(), Trials = sum(Trials),
              Accuracy = mean(Accuracy), .groups = "drop")
  kable(caf_summary, digits = 3,
        caption = "Contributing counts and mean participant-bin accuracy (0-1 scale).")
  p_conditional_accuracy <- ggplot(
    caf_summary, aes(Quintile, Accuracy, colour = Condition, shape = Condition)
  ) +
    geom_line(aes(group = Condition)) + geom_point(size = 2.5) +
    scale_colour_manual(values = condition_colours, labels = condition_labels) +
    scale_shape_manual(values = c(A = 16, B = 17), labels = condition_labels) +
    scale_x_continuous(breaks = 1:5, labels = c("1 (fastest)", "2", "3", "4", "5 (slowest)")) +
    scale_y_continuous(limits = c(0, 1), labels = scales::label_percent()) +
    labs(x = "Within-participant RT quintile", y = "Mean participant-bin accuracy",
         title = "Accuracy and relative response speed") + theme(legend.position = "bottom")
  print(p_conditional_accuracy)
} else {
  cat("No participant-condition combinations have enough eligible trials for five bins.")
}
```

Equal-frequency bins can split tied RTs; with ties, bin membership also depends on their ordering. A declining curve would describe lower accuracy among slower responses in these data. Neither a declining nor an increasing curve establishes what would happen if a participant were instructed to respond faster. In this simulator, RT is generated conditional on correctness.

Verify one calculation directly

Select one participant–condition combination with an eligible RT and an observed correctness score. Recompute its median and accuracy from the raw import using base R. Agreement checks one concrete case; it does not validate all the analysis choices.

```
check_row <- person %>%
  filter(N_RT > 0, N_scored > 0) %>%
  arrange(Subject, Condition) %>%
  slice_head(n = 1)

if (nrow(check_row) == 1) {
  chosen_subject <- as.character(check_row$Subject)
  chosen_condition <- as.character(check_row$Condition)
  one <- raw[raw$Subject == chosen_subject &
               unname(condition_map[as.character(raw$Condition)]) == chosen_condition, ]
  scores <- one$Correct[!is.na(one$Correct)]
  manual_accuracy <- length(scores[scores == 1]) / length(scores)
  keep <- which(is.finite(one$RT) & one$RT > 0 & one$Correct == 1)
  manual_median <- median(one$RT[keep])

  stopifnot(
```

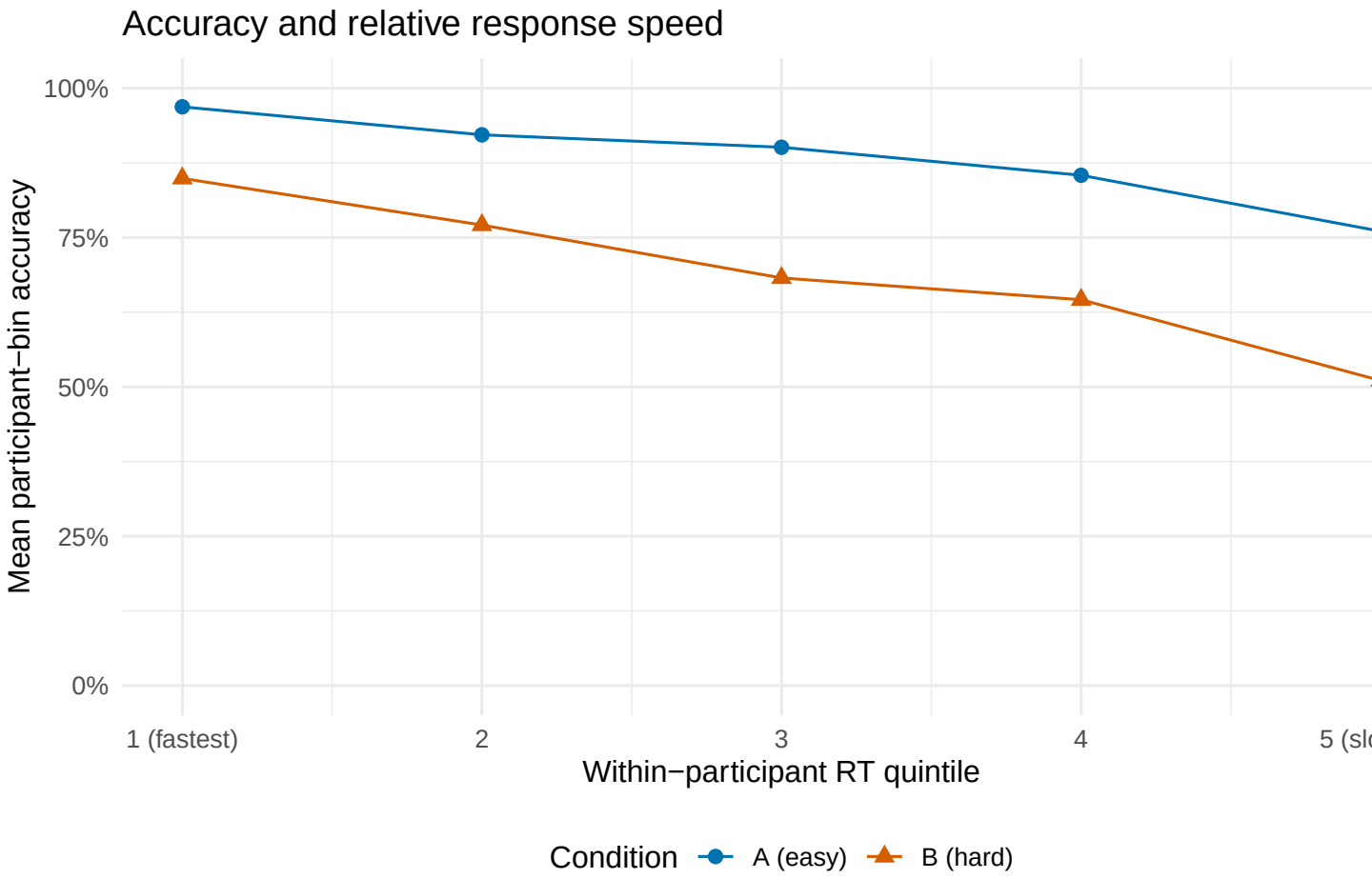


Figure 9: Mean participant-bin accuracy by within-participant RT rank quintile. Only positive finite RTs with observed correctness enter this supplementary analysis. It uses a different denominator from overall accuracy.

```
    isTRUE(all.equal(manual_accuracy, check_row$Accuracy)),
    isTRUE(all.equal(manual_median, check_row$Median_RT))
  )
  kable(tibble(
    Subject = chosen_subject, Condition = chosen_condition,
    Median_RT_ms = manual_median, Accuracy = manual_accuracy,
    Verification = "Agrees with participant summary"
  ), digits = 3, caption = "A direct check against the raw observations.")
} else {
  cat("No participant-condition combination supports both verification calculations.")
}
```

Table 11: A direct check against the raw observations.

Subject	Condition	Median_RT_ms	Accuracy	Verification
S01	A	486.041	0.875	Agrees with participant summary

Interpretation and limitations

Use the computed results to describe the magnitude and direction of the paired differences, how many participants follow each direction, and how much their responses vary. Read those differences alongside the trial distributions and both performance outcomes. A faster response alone does not establish better performance.

The main RT analysis is conditional on correct responses and therefore does not describe the timing of all responses. The all-response comparison helps assess the consequences of that selection, while the accuracy analysis describes another aspect of performance. Selective missingness, unequal eligible trial counts, and any extreme recorded RTs remain relevant limitations. Participant medians based on fewer trials are less stable; the equal participant weighting used here does not account for that difference in precision.

The summaries are descriptive. No formal test or causal effect is claimed, and apparent differences do not establish a speed–accuracy mechanism. In simulation mode these findings describe a constructed teaching dataset, and their broad direction is built into the generating model. The report does not assess practice, fatigue, or post-error slowing: the generator has no sequential mechanism and provides no verified global presentation order. In CSV mode, additional information about recruitment and the actual study protocol would be needed for broader interpretation.

Comparison with the previous solutions

Feature or issue	Decision in this report
The earlier worked response required an unavailable CSV and accepted only A/B	Added an explicit simulation/CSV choice and a checked mapping for A/B and A (Easy)/B (Hard).
The attachment contains the full simulator	Retained its random draws, row ordering, sample size, and parameters; made the RNG settings explicit. Corrected the accuracy random effect description to log-odds.
Descriptive summaries, paired plots, and distribution views	Retained these ideas; made outcome-specific denominators and complete pair counts visible.

Feature or issue	Decision in this report
`Pct = n()/nrow(sim_data)` in the attachment's summary	The added response table uses the within-condition denominator and names it explicitly. The original calculation describes a fraction of all trials.
Correct/incorrect RT densities	Retained, stratified by condition, using a shared bandwidth and distinguishing distributional shape from response frequency.
Raincloud plots	Used boxplots with horizontal jitter and no vertical jitter, plus the density view. Accuracy is bounded and discrete here; a smooth density can suggest unsupported detail.
Joint RT–accuracy view	Retained matched participant movements, removed the claim that the plot proves a trade-off, and added a plot of paired changes.
`stat_ellipse(level = 0.80)` labelled as an 80% CI	Removed. A data ellipse describes scatter under a distributional model; it is not an 80% confidence region for the group mean.
Trial-index LOESS and post-error analysis	Omitted from substantive conclusions. The simulator has no learning/fatigue/previous-error term, and condition-specific indices do not establish a global trial order. Naive pooled smooth bands also ignore participant clustering.
Heatmaps and quantile differences	Retained as complementary descriptive views, with shared participant ordering and paired quantile calculations.
Conditional accuracy by RT quintile	Retained as an optional association plot, with participant-level bin summaries before averaging and an explicit minimum of five scored valid-time trials.
Mixed models, paired tests, and generic effect-size labels	Left out of this descriptive activity. A t-test remains inferential even when called descriptive. Any later inference needs its own assumptions and diagnostics.
Log-RT back-transformation in the attachment	Exponentiated fitted log values describe conditional medians/geometric means under the model, not automatically arithmetic mean RTs. Interaction contrasts also depend on the reference condition/correctness.
Repeated dashboards, reused plot names, duplicated text, dangling fence	Replaced with individual named plots and one report structure.

The previous correct-versus-incorrect comparison pooled across conditions in some appendices. Since condition is associated with both accuracy and RT, that pooled contrast also reflects a different mixture of conditions. The corresponding plots and tables here keep condition visible.

Documentation supporting the graphical corrections: ggplot2 data ellipses and ggplot2 smooths.

> **AI assistance.** This report source was prepared with ChatGPT by comparing two earlier teaching examples. Its numerical logic was checked using a Python port of the supplied simulator and independent calculations. Native R execution and knitting were unavailable in the preparation environment; this limitation should not be mistaken for a successful R runtime test. The final interpretation remains the author's responsibility.

```r
sessionInfo()
```

```
## R version 4.6.1 (2026-06-24 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26200)
##
## Matrix products: default
##   LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=Italian_Italy.utf8  LC_CTYPE=Italian_Italy.utf8
## [3] LC_MONETARY=Italian_Italy.utf8 LC_NUMERIC=C
## [5] LC_TIME=Italian_Italy.utf8
##
## time zone: Europe/Rome
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] knitr_1.51      patchwork_1.3.2 scales_1.4.0    lubridate_1.9.5 forcats_1.0.1
##  [6] stringr_1.6.0   dplyr_1.2.1     purrr_1.2.2     readr_2.2.0     tidyr_1.3.2
## [11] tibble_3.3.1    ggplot2_4.0.3   tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
##  [1] ggfittext_0.10.4     gtable_0.3.6         beeswarm_0.4.0       circlize_0.4.18
##  [5] shape_1.4.6.1        xfun_0.60            htmlwidgets_1.6.4    GlobalOptions_0.1.4
##  [9] lattice_0.22-9       GGally_2.4.0         tzdb_0.5.0           crosstalk_1.2.2
## [13] vctrs_0.7.3          tools_4.6.1          generics_0.1.4       parallel_4.6.1
## [17] pkgconfig_2.0.3      Matrix_1.7-5         data.table_1.18.6.1  RColorBrewer_1.1-3
## [21] S7_0.2.2             ggridges_0.5.7       distributional_0.8.1 lifecycle_1.0.5
## [25] compiler_4.6.1       farver_2.1.2         progress_1.2.3       tinytex_0.60
## [29] vipor_0.4.7          htmltools_0.5.9      yaml_2.3.12          plotly_4.12.1
## [33] crayon_1.5.3         pillar_1.11.1        MASS_7.3-65          nlme_3.1-169
## [37] ggstats_0.13.0       tidyselect_1.2.1     digest_0.6.39        stringi_1.8.9
## [41] labeling_0.4.3       splines_4.6.1        fastmap_1.2.0        grid_4.6.1
## [45] colorspace_2.1-3     treemapify_2.6.1     cli_3.6.6            magrittr_2.0.5
## [49] withr_3.0.3          prettyunits_1.2.0    bit64_4.8.4          ggbeeswarm_0.7.3
## [53] timechange_0.4.0     rmarkdown_2.31       httr_1.4.8           bit_4.6.0
## [57] otel_0.2.0           hms_1.1.4            ggalluvial_0.12.6    evaluate_1.0.5
## [61] ggdist_3.3.3         viridisLite_0.4.3    mgcv_1.9-4           rlang_1.3.0
## [65] isoband_0.3.0        Rcpp_1.1.2           glue_1.8.1           vroom_1.7.1
## [69] rstudioapi_0.19.0    jsonlite_2.0.0       R6_2.6.1
```